

American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

Mid-Term Examination Suggestions

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Answer all questions.

Total Marks: 30

(20 × 1.5 = 30)

1. During software maintenance, developers modify an application so that it can run on a newly released operating system without changing its original functionality. Which type of maintenance is this?

- A. Corrective Maintenance
- B. Adaptive Maintenance
- C. Perfective Maintenance
- D. Preventive Maintenance

Answer: B

2. According to Software Engineering principles, the primary objective of the Software Development Life Cycle (SDLC) is to:

- A. Increase programming speed only.
- B. Develop software through a structured process.
- C. Eliminate software documentation.
- D. Replace software testing completely.

Answer: B

3. Which software process model associates every development activity with a corresponding testing activity?

- A. Agile Model
- B. Prototype Model
- C. V-Model
- D. Spiral Model

Answer: C

4. Which software process model develops software quickly by constructing modules in parallel and integrating them later?

- A. Waterfall Model
- B. RAD Model
- C. V-Model
- D. Prototype Model

Answer: B

5. A banking application must encrypt all customer information using AES-256. This requirement is classified as:

- A. Functional Requirement
- B. Business Requirement
- C. Non-functional Requirement
- D. User Requirement

Answer: C

6. Which Requirements Engineering phase is responsible for collecting requirements from customers through interviews, surveys, and workshops?

- A. Validation
- B. Elicitation
- C. Specification
- D. Negotiation

Answer: B

7. Which of the following is the BEST example of scope creep?

- A. Developers fix coding errors.
- B. New features are continuously added without formal approval.
- C. Test cases are reviewed.
- D. Requirements are documented in SRS.

Answer: B

8. Agile development is particularly suitable for projects where:

- A. Requirements never change.
- B. Customer feedback is frequent.
- C. Documentation is the highest priority.
- D. Testing is performed only after coding.

Answer: B

9. Agile teams usually prefer shorter iterations because they:

- A. Reduce customer involvement.
- B. Increase project risk.
- C. Provide faster feedback and lower complexity.
- D. Eliminate software testing.

Answer: C

10. Which XP practice requires two programmers to work together on a single computer?

- A. Test-Driven Development
- B. Continuous Integration
- C. Pair Programming
- D. Refactoring

Answer: C

11. In Extreme Programming (XP), the collection of user stories used for implementation planning is prepared mainly during which phase?

- A. Exploration Phase
- B. Planning Phase
- C. Maintenance Phase
- D. Death Phase

Answer: B

12. Which Scrum artifact contains all known and prioritized product requirements?

- A. Sprint Backlog
- B. Product Increment
- C. Product Backlog
- D. Burndown Chart

Answer: C

13. Which Scrum phase prepares the completed system for release through integration, documentation, and system testing?

- A. Pre-game Phase
- B. Development Phase
- C. Post-game Phase
- D. Sprint Planning

Answer: C

14. In DSDM, which activity determines whether DSDM is appropriate for a project?

- A. Business Study
- B. Feasibility Study
- C. Functional Iteration
- D. Implementation

Answer: B

15. Which DSDM technique uses fixed deadlines while allowing lower-priority requirements to be postponed?

- A. Pair Programming
- B. Code Review
- C. Timeboxing
- D. Continuous Delivery

Answer: C

16. Which FDD process assigns feature sets to Chief Programmers and determines class owners?

- A. Build by Feature
- B. Plan by Feature
- C. Develop Overall Model
- D. Design by Feature

Answer: B

17. In Feature Driven Development (FDD), most individual features are expected to be completed within:

- A. One day
- B. One month
- C. One to five days
- D. Six months

Answer: C

18. Which management myth suggests that adding additional programmers will always recover a delayed project schedule?

- A. Software is easy to maintain.
- B. Modern tools solve every problem.
- C. More programmers always speed up development.
- D. Documentation is unnecessary.

Answer: C

19. Which software failure commonly occurs when project requirements are misunderstood during the early stages of development?
- A. Faster software delivery
 - B. Improved user satisfaction
 - C. Cost overruns and project failure
 - D. Reduced maintenance effort

Answer: C

20. Which practice encourages verification of software requirements before coding begins to reduce the cost of defects?
- A. Continuous Deployment
 - B. Shift-Left Testing
 - C. Pair Programming
 - D. Sprint Retrospective

Answer: B